

Case Study: 15-04-26

Product Management with Image Upload

1. Resource Group Creation
2. Virtual Network (VNet)

The screenshot shows the Azure portal interface for the 'project-vnet' resource group. The left sidebar contains navigation links: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Monitoring, Automation, and Help. The main pane displays the 'Essentials' section with the following details:

- Resource group (move): rg-project
- Location (move): Central India
- Subscription (move): Azure for Students
- Subscription ID: 1fc5191b-5591-4f01-bae6-9b80cdd19b3a
- Address space: 10.0.0.0/16
- Subnets: 2 subnets
- DNS servers: Azure provided DNS service
- BGP community string: Configure
- Virtual network ID: 97e71bf6-2af4-443b-a608-14f07cb44a49

Tags (edit) and Add tags links are at the bottom. A 'JSON View' link is in the top right. A 'Notifications' pane on the right shows three events: 'Successfully updated server firewall rules', 'Successfully created storage container', and 'Successfully updated storage account', all occurring 'an hour ago'. The Windows taskbar at the bottom shows the date as 15-04-2026 and time as 14:12.

3. Subnet Configuration

The screenshot shows the 'project-vnet | Subnets' configuration page in the Azure portal. The left sidebar is the same as the previous screenshot. The main pane displays instructions: 'Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.' Below this is a 'Search subnets' input field and a table of subnets:

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Sec
☐	frontend-s...	10.0.1.0/24	-	250	-	-
☐	backend-s...	10.0.2.0/24	-	250	-	-

At the top of the main pane, there are links for '+ Subnet', 'Refresh', 'Manage users', 'Delete', and 'Export to CSV'. The Windows taskbar at the bottom shows the date as 15-04-2026 and time as 14:13.

4. Frontend Network Security Group (NSG)

Frontend-nsg - Microsoft Azure

portal.azure.com/#@sakshamchauhanf111gmail.onmicrosoft.com/resource/subscriptions/1fc5191b-5591-4f01-bae6-9b80cdd19b...

Frontend-nsg

Network security group

Location: Central India

Subscription ID: 1fc5191b-5591-4f01-bae6-9b80cdd19b3a

Tags (edit) Add tags

Filter by name

Port == all Protocol == all Source == all Destination == all

Priority	Name	Port	Protocol	Source	Destination
100	HTTP	80	TCP	Any	Any
110	HTTPS	443	TCP	Any	Any
120	RDP	3389	TCP	Any	Any
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork

Very hot weather Now

5. Backend Network Security Group

Backend-nsg - Microsoft Azure

portal.azure.com/#@sakshamchauhanf111gmail.onmicrosoft.com/resource/subscriptions/1fc5191b-5591-4f01-bae6-9b80cdd19b...

Backend-nsg

Network security group

Subscription ID: 1fc5191b-5591-4f01-bae6-9b80cdd19b3a

Tags (edit) Add tags

Filter by name

Port == all Protocol == all Source == all Destination == all

Priority	Name	Port	Protocol	Source	Destination
100	allow-API	5000	TCP	10.0.1.0/24	Any
110	allow-RDP	3389	TCP	115.244.89.211	Any
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork
65001	AllowAzureLoadBalan...	Any	Any	AzureLoadBalancer	Any
65500	DenyAllInBound	Any	Any	Any	Any

Air: Moderate Now

6. Frontend Virtual Machine

The screenshot displays the Azure portal interface for a virtual machine named 'Frontend-VM'. The left-hand navigation pane includes sections for Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Monitor, Resource visualizer, Connect, and Networking. The main content area shows an Advisor alert at the top: 'Windows virtual machines should enable Azure Disk Encryption or EncryptionAtHost'. Below this, there are buttons for 'Help me copy this VM in any region' and 'Manage this VM with Azure CLI'. A row of action buttons includes Connect, Start, Restart, Stop, Hibernate, Capture, Delete, Refresh, Scale, and Open in monitor. The VM's details are listed in two columns: Resource group (rg-project), Status (Stopped (deallocated)), Location (Central India (Zone 2)), Subscription (Azure for Students), Subscription ID (1fc5191b-5591-4f01-bae6-9b80cdd19b3a), Availability zone (2), Operating system (Windows), Size (Standard B2ats v2 (2 vcpus, 1 GiB memory)), Primary NIC public IP (20.198.24.78), Virtual network/subnet (project-vnet/frontend-subnet), DNS name (Not configured), and Health state (-).

7. Backend Virtual Machine

The screenshot displays the Azure portal interface for a virtual machine named 'Backend-VM'. The layout is similar to the Frontend-VM page, with a left navigation pane and a main content area. The Advisor alert at the top states: 'Windows servers should be configured to use secure communication protocols'. The action buttons and VM details are consistent with the previous page. The details listed are: Resource group (rg-project), Status (Stopped (deallocated)), Location (Central India (Zone 2)), Subscription (Azure for Students), Subscription ID (1fc5191b-5591-4f01-bae6-9b80cdd19b3a), Availability zone (2), Operating system (Windows), Size (Standard B2ats v2 (2 vcpus, 1 GiB memory)), Primary NIC public IP (20.198.24.78), Virtual network/subnet (project-vnet/backend-subnet), DNS name (Not configured), and Health state (-).

8. Networking Between VMs

smartretail - Microsoft Azure | Product Management API | Swagger UI | Ask Gemini

portal.azure.com/#@sakshamchauhanf1111gmail.onmicrosoft.com/resource/subscriptions/1fc5191b-5591-4f01-bae6-9b80cdd19b...

Home > rg-project > smartretail

smartretail | Networking

SQL server

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Quick start
- Diagnose and solve problems
- Resource visualizer
- Settings
 - Microsoft Entra ID
 - SQL databases
 - SQL elastic pools
 - Properties

Rule	Virtual network	Subnet	Address range	Endpoint status	Resource group	Subscription	State
Firewall rules Allow certain public internet IP addresses to access your resource. Learn more							
+ Add your client IPv4 address (49.249.148.158) + Add a firewall rule							
Rule name		Start IPv4 address	End IPv4 address				
ClientIPAddress_2026-4-15_12-48-53		49.249.148.158	49.249.148.158				
Exceptions							
☒ Allow Azure services and resources to access this server							
Save Discard							

36°C Sunny 14:19 15-04-2026

9. Azure SQL Database

SmartRetailDB (smartretail/Sm... | Product Management API | Swagger UI | Ask Gemini

portal.azure.com/#@sakshamchauhanf1111gmail.onmicrosoft.com/resource/subscriptions/1fc5191b-5591-4f01-bae6-9b80cdd19b...

Home

SmartRetailDB (smartretail/SmartRetailDB)

SQL database

Search resources, services, and docs (G+/I) | Copilot | sakshamchauhanf1111... | DEFAULT DIRECTORY (SAKSHAM...)

Summarize the properties for this SQL database. +2

Copy Restore Export Set server firewall Delete Open in Feedback

Overview

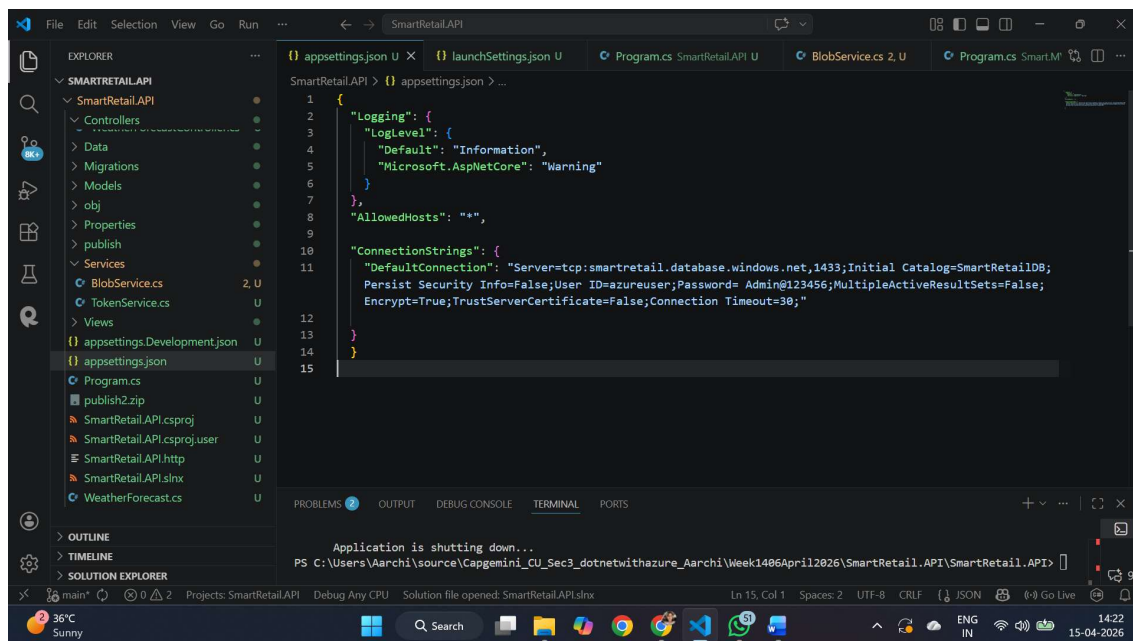
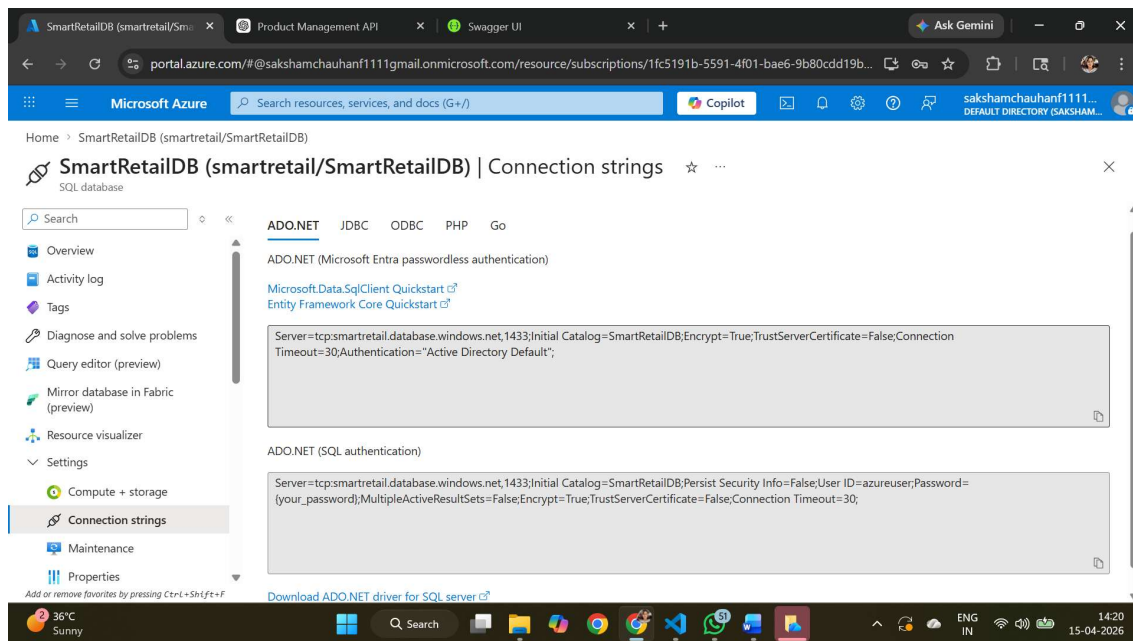
- Activity log
- Tags
- Diagnose and solve problems
- Query editor (preview)
- Mirror database in Fabric (preview)
- Resource visualizer
- Settings
 - Compute + storage
 - Connection strings
 - Maintenance
 - Properties

Essentials

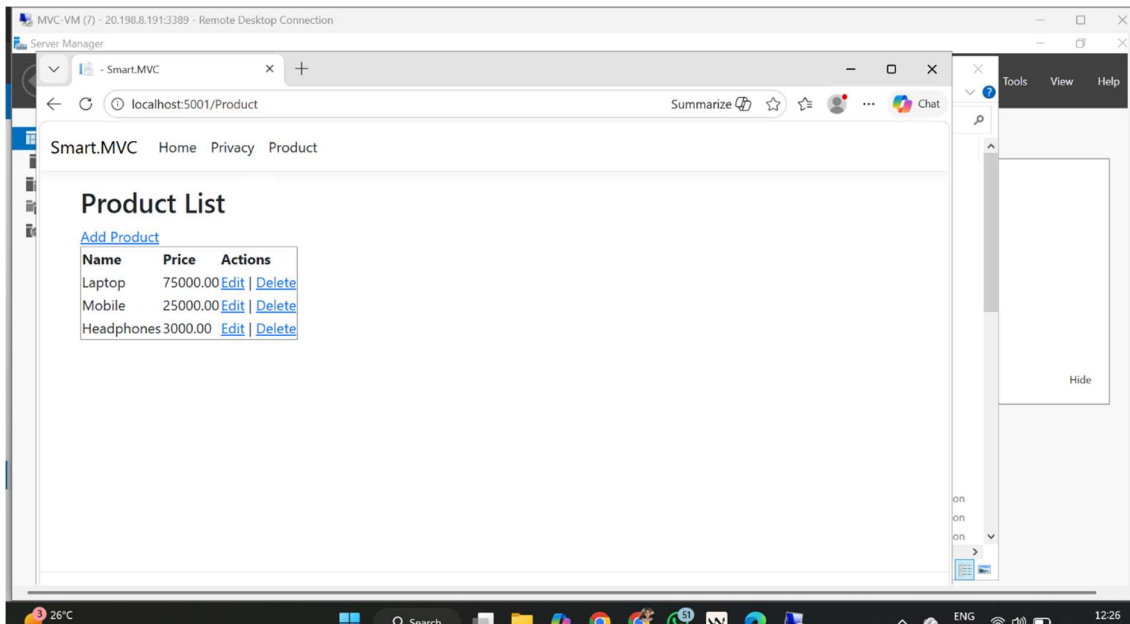
Resource group (move)	Server name
rg-project	smartretail.database.windows.net
Status	Connection strings
Resuming	Show database connection strings
Location	Pricing tier
Central India	Free - General Purpose - Serverless: Gen5, 2 vCores
Subscription (move)	Overage billing
Azure for Students	Disabled
Subscription ID	Free monthly vCore amount
1fc5191b-5591-4f01-bae6-9b80cdd19b3a	96,728 vCore seconds remaining
Tags (edit)	Earliest restore point
Add tags	2026-04-13 10:25 UTC

Add or remove favorites by pressing Ctrl+L+Sn (L+Sn)

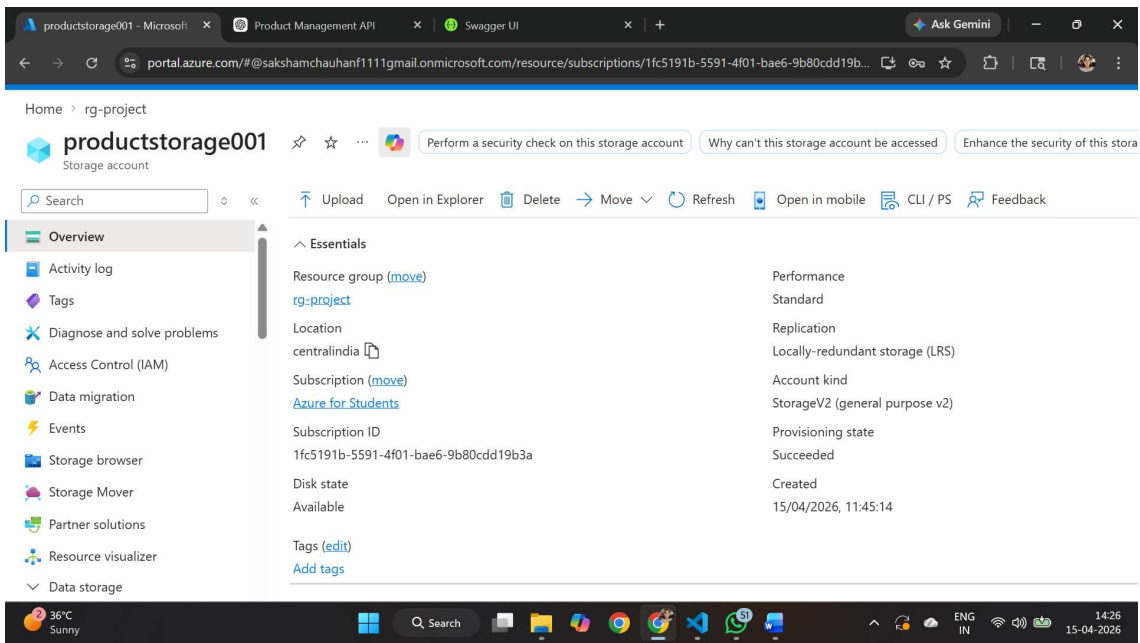
36°C Sunny 14:20 15-04-2026

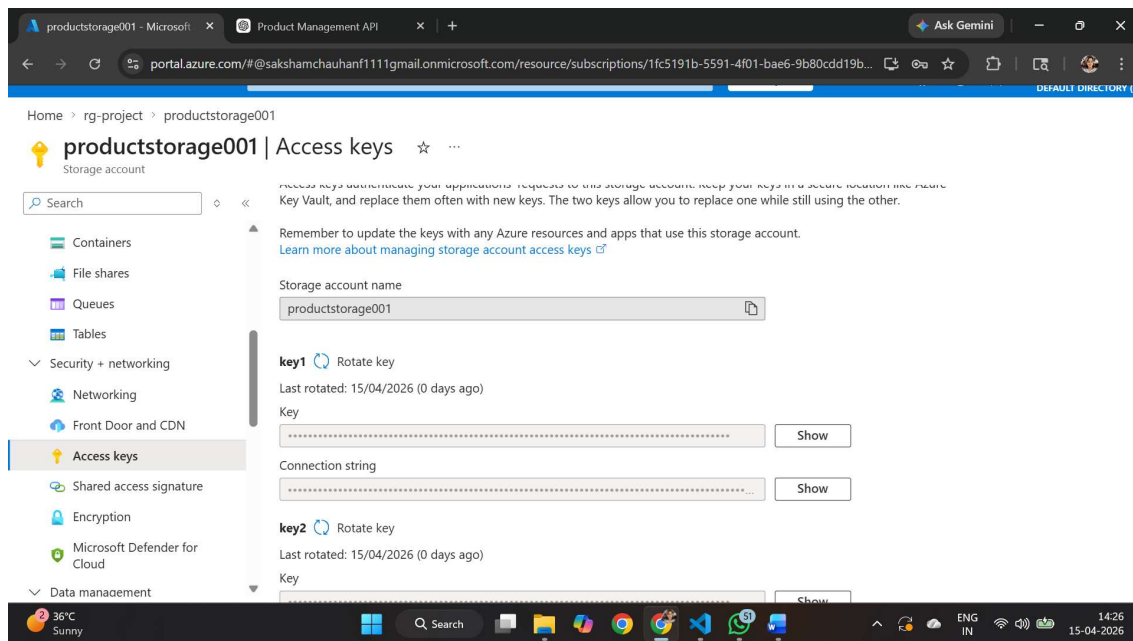


10. MVC on VM

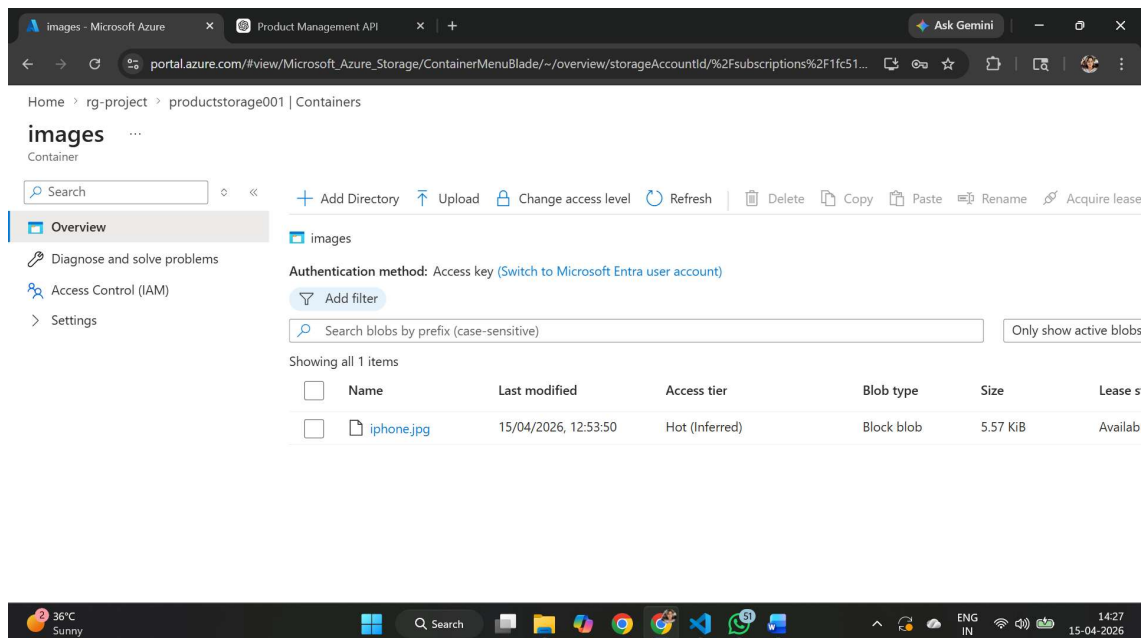


11. Storage Account Creation





12. Blob Container Creation



13. Azure Key Vault Integration

KeyVault15 - Microsoft Azure x Product Management API x + Ask Gemini

portal.azure.com/#@sakshamchauhanf1111gmail.onmicrosoft.com/resource/subscriptions/1fc5191b-5591-4f01-bae6-9b80cdd19b...

Home > rg-project

KeyVault15

Key vault

Show me total service API hits metrics of this Key Vault. What is the overall service API latency for this Key Vault? +1

Search

Delete Move Refresh Open in mobile

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Access policies
- Resource visualizer
- Events
- Objects
 - Keys
 - Secrets

Essentials

Resource group (move)
[rg-project](#)

Location
Central India

Subscription (move)
[Azure for Students](#)

Subscription ID
1fc5191b-5591-4f01-bae6-9b80cdd19b3a

Vault URI
<https://keyvault15.vault.azure.net/>

Sku (Pricing tier)
Standard

Directory ID
6e3b9983-05eb-4039-bc18-d014f5b66b17

Directory Name
Default Directory

Soft-delete
[Enabled](#)

Purge protection
[Disabled](#)

Tags (edit)
[Add tags](#)

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KeyVault15 - Microsoft Azure x Product Management API x + Ask Gemini

portal.azure.com/#@sakshamchauhanf1111gmail.onmicrosoft.com/resource/subscriptions/1fc5191b-5591-4f01-bae6-9b80cdd19b...

Home > rg-project > KeyVault15

KeyVault15 | Secrets

Key vault

Search

+ Generate/Import Refresh Restore Backup Manage deleted secrets View sample code

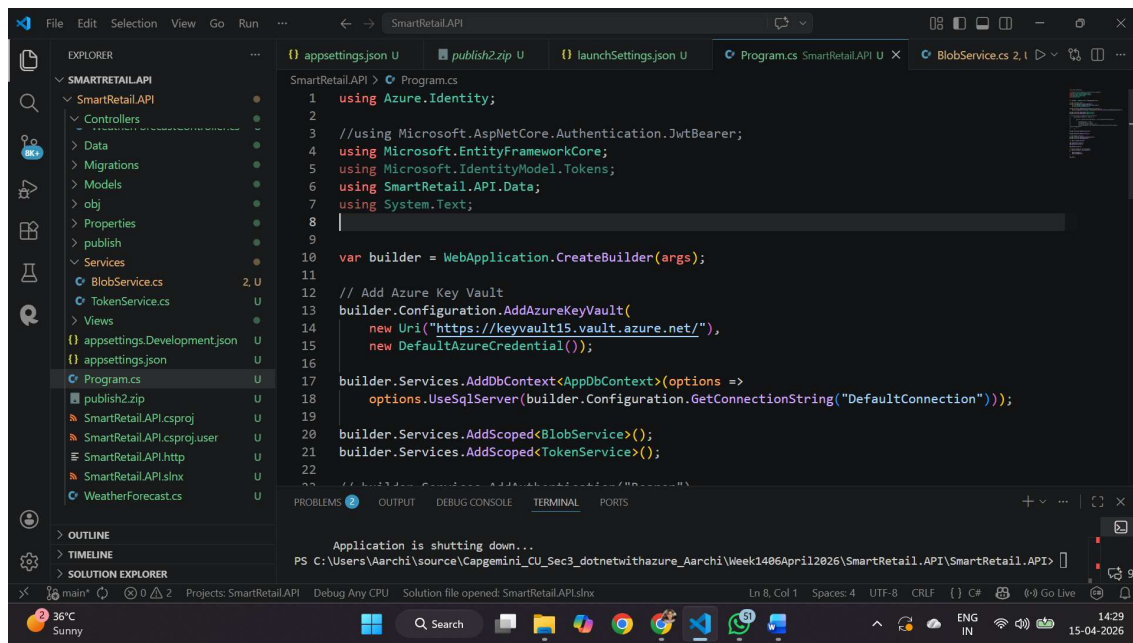
Name	Type	Status	Expiration date
BlobConnection		✓ Enabled	

Overview

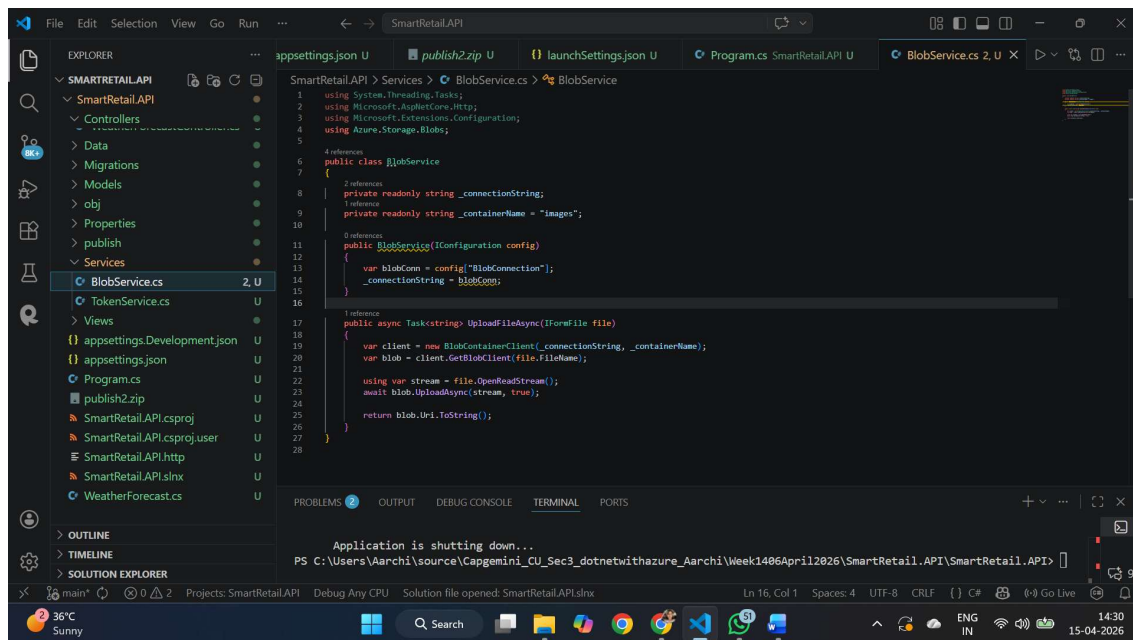
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Access policies
- Resource visualizer
- Events
- Objects
 - Keys
 - Secrets
 - Certificates

36°C Sunny 14:28 15-04-2026

14. Secure Configuration in API



15. Blob Service Implementation



16. Database Integration

SmartRetailDB (smartretail/SmartRetailDB) | Query editor (preview)

SQL database

Search

New query Templates Open in Feedback Sign out Classic experience

Overview
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SmartRetailDB

dbo

Queries

SQL query 1

SQL query 1

Run Save as view

Your queries will be lost when your session ends. Use **Save as view** to keep results as a database view, or copy your query text to save it externally.

```
1 ALTER TABLE Products
2 ADD ImageUrl NVARCHAR(MAX);
```

SmartRetailDB (smartretail/SmartRetailDB) | Query editor (preview)

SQL database

Search

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SmartRetailDB

dbo

Queries

SQL query 1

SQL query 1

Run Save as view

Your queries will be lost when your session ends. Use **Save as view** to keep results as a database view, or copy your query text to save it externally.

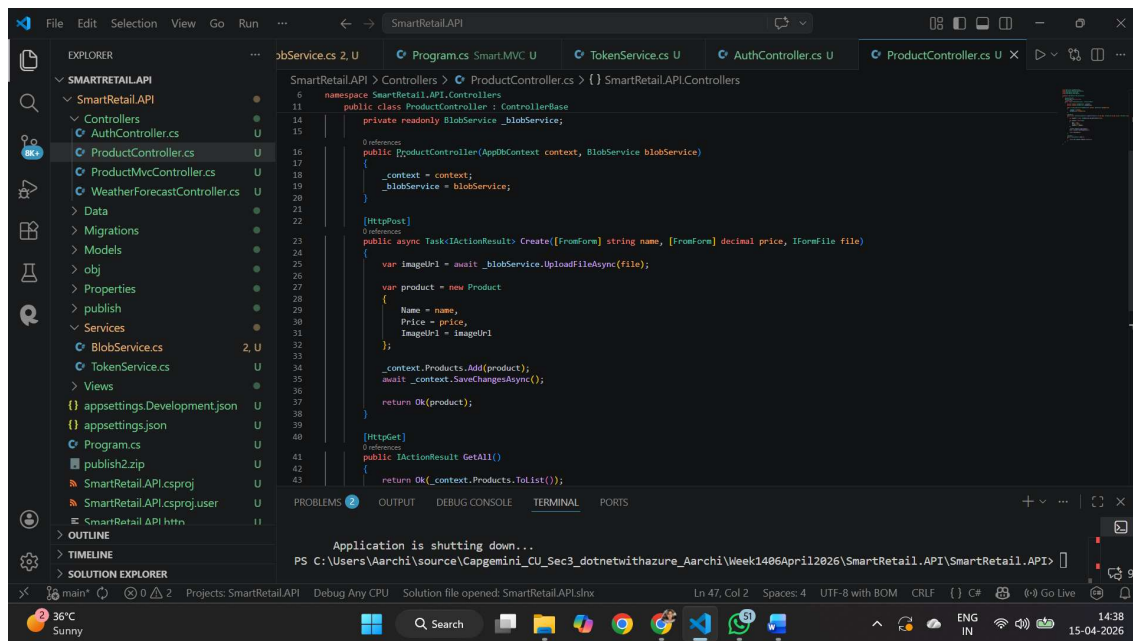
```
1 SELECT * FROM Products;
```

Messages Results

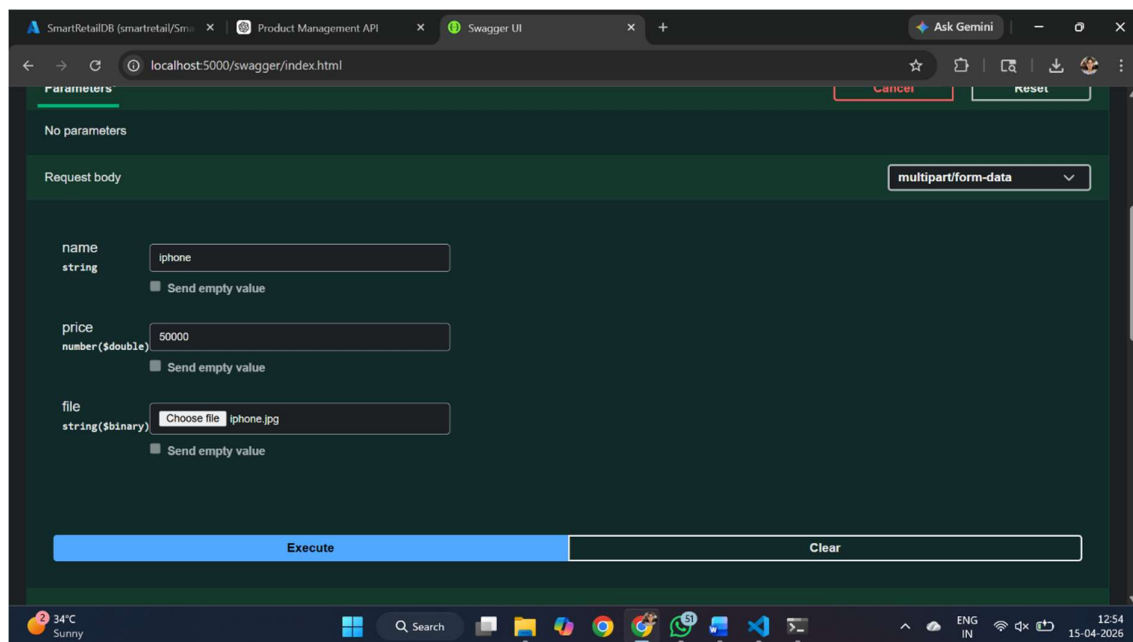
	Id	ABC Name	Price	ABC ImageUrl
1	1	Laptop	75000.00	NULL
2	2	Mobile	25000.00	NULL
3	3	Headphones	3000.00	NULL

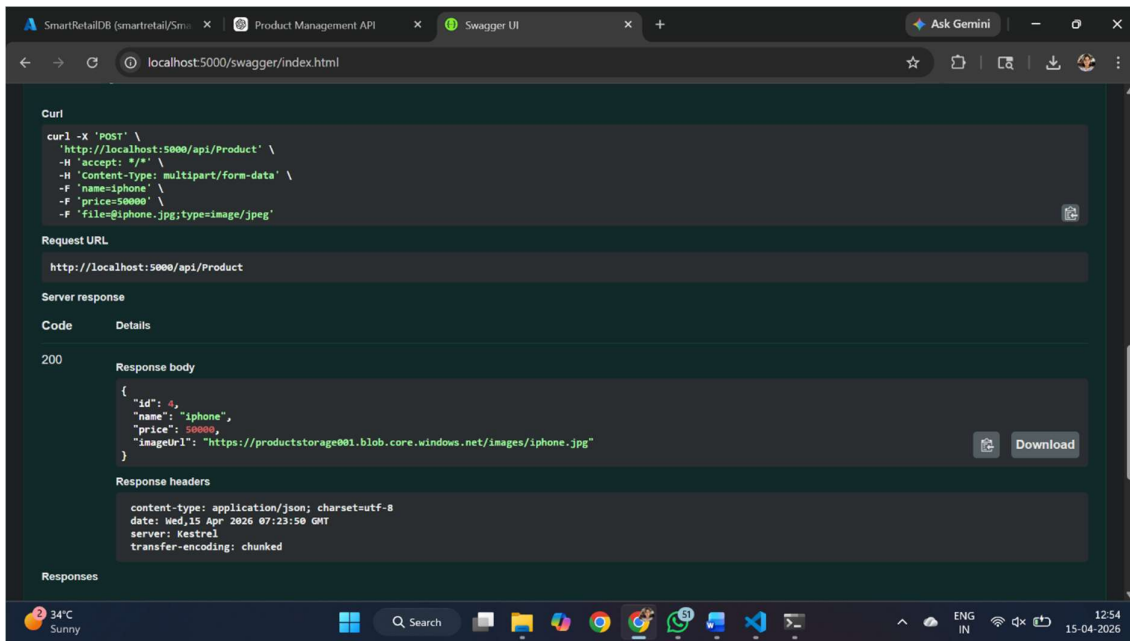
Succeeded (0 sec 168 ms)

17. API Development



18. Tested APIs using Swagger





19. Checking Image URL in Azure

